Computers and Education: Artificial Intelligence 8 (2025) 100380

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence

Enhancing data analysis and programming skills through structured prompt training: The impact of generative AI in engineering education

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ARTICLE INFO

Keywords:
Applications in subject areas
Post-secondary education
Teaching/learning strategies

ABSTRACT

The advent of Generative Artificial Intelligence (GenAI) and large language models like LLama, Palm2, GPT, Gemini, and Claude has revolutionized education by generating human-like text and contextually relevant responses. Our research investigates the impact of structured prompt training on students' learning in data analysis and programming. We experimented with 157 first-year engineering students divided into three groups: a control group (internet access, no GenAI), an experimental group 1 (internet and GenAI without prompt training), and an experimental group 2 (internet and GenAI with prompt training). The prompt training session included techniques like few-shot prompting, chain prompting, and the CLEAR framework. We assessed participants' performance in data analysis tasks using Python, with pre-tests and post-tests measuring their skills in programming across three Bloom's taxonomy levels (understanding, application, and analysis). ANOVA on post-test scores showed significant differences among the groups, with G3 (with prompt training) outperforming G2 (without prompt training) and the control group across all three levels, evidenced by higher mean scores (G3: 6.60, G2: 4.94, Control: 4.28), similar pattern observed in task completion also. These results underscore the effectiveness of structured prompt training in enhancing students' data analysis and programming skills. Our study highlights the potential of GenAI and structured prompt training to transform educational practices and suggests future research directions, including integrating prompt engineering within human-AI collaboration.

1. Introduction

In recent years, the advent of Generative artificial intelligence (GenAI) and large language models (LLMs)[1] has revolutionized various fields like healthcare, business, and education (Gozalo-Brizuela & Garrido-Merchan, 2023) (Bozkurt, 2023). These models can generate human-like text and provide instant, contextually relevant responses. As such, they have become valuable tools for students and educators, offering support in various educational activities like writing assistance, providing supplementary learning resources, guiding in learning programming languages, and support for instructors in evaluating, content generation (Rahman & Watanobe, 2023) (Michel-Villarreal et al., 2023) (Steiss et al., 2024). Initially, GenAI tools were primarily used for text generation. However, with the advancements in natural language processing (NLP) and machine learning, GenAI has transitioned into more interactive roles such as creating personalized learning experiences, providing feedback, tutoring students in specific subjects, and generating various modalities such as audio, image, video, etc (Gozalo-Brizuela & Garrido-Merchan, 2023) (Bozkurt, 2023). These capabilities have opened new avenues for enhancing educational outcomes and addressing diverse learning needs.

The benefits of incorporating GenAI in educational settings are manifold. Firstly, they offer on-demand assistance, allowing students to access help whenever needed, which is particularly beneficial in self-paced learning environments (Cao & Dede, 2023) (Seo et al., 2021). Secondly, these models can cater to different learning styles by providing explanations in multiple formats, such as text, summaries, or even step-by-step guides, and have been shown to improve the performance and engagement of students (Pesovski et al., 2024). Thirdly, GenAI can assist educators by automating repetitive tasks like evaluating assessments, creating lesson plans, etc., thus freeing up time for more personalized instruction and interaction with students (Zhai, 2023) (Khazanchi & Khazanchi, 2024) (Schroeder et al., 2022).

Despite these advantages, there are notable challenges and limitations associated with the use of LLMs in education. One of the primary concerns is the quality and accuracy of the generated content. LLMs

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[1] LLaMA, PaLM2, GPT-3, and Claude 2.

https://doi.org/10.1016/j.caeai.2025.100380
Received 1 October 2024; Received in revised form 10 January 2025; Accepted 3 February 2025
Available online 10 February 2025
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are trained on vast datasets that include both accurate information and misinformation, which can sometimes lead to the generation of incorrect or biased responses (Doughty et al., 2024). Additionally, the lack of contextual understanding often results in generic or irrelevant information (Siontis et al., 2024) (Weisz et al., 2024) (Michel-Villarreal et al., 2023). Moreover, the lack of skill in how to effectively use these tools can result in suboptimal outcomes (Yu & Guo, 2023) (Barbas et al., 2023). To minimize these challenges, one way is to fine-tune AI models with improved datasets and feedback ensuring greater accuracy and relevance (Dave et al., 2023). Another technique suggested is prompt engineering, which involves crafting precise and context-specific queries to guide AI responses (Park et al., 2023) (Alier et al., 2024). Studies show that prompting methods like zero-shot and few-shot prompting allow models to handle new tasks with minimal training data, leveraging pre-existing knowledge for accurate responses (Radford et al., 2019) (Brown et al., 2020). Another prompting method, Chain-of-Thought (CoT) prompting improves reasoning by breaking down tasks into step-by-step processes, enhancing accuracy in complex problem-solving (Wei et al., 2022) (Zhang et al., 2022). In educational contexts, prompt training could empower students to better leverage AI tools for learning, leading to improved academic performance and a deeper understanding of the subject matter (Walter, 2024). However, there are limited studies analyzing the effects of prompting skills and student performance, suggesting a need for further research in this area.

Building on this understanding, our study considers prompt training as an intervention to assess its effect on the learning of students in the domain of data analysis and programming. Recent research in data science indicates that proficiency in data analysis skills enables individuals to effectively interpret complex datasets and uncover meaningful patterns (Donoho, 2024) (Nasution et al., 2023). For example, programming languages like Python automate data processing and facilitate reproducible analyses, enhancing efficiency and accuracy (Rogel-Salazar, 2023). Moreover, the job market increasingly demands these competencies, with employers seeking professionals who can transform raw data into actionable insights (Donoho, 2024) (Nasution et al., 2023).

To study the impact of prompt training, we designed an experiment in the domain of data analysis using prompt training with ChatGPT. Our study involves three groups of participants:

- Control Group: Participants have access to the internet but do not have access to generative AI tools. (G1)
- Experimental Group 1: Participants have access to the internet and GenAI (ChatGPT) without any prompt training. (G2)
- Experimental Group 2: Participants have access to the internet and GenAI (ChatGPT) with prompt training. (G3)

All the groups were given a three-hour task to measure the data analysis skills focusing on two basic concepts - data aggregation and merging. While experimental group 1 had access to internet, experimental group 2 underwent an hour-long training session to learn essential prompting skills while using ChatGPT. This session educates students on prompting techniques such as few-shot prompting, chain prompting, and the CLEAR framework of prompting (Lo, 2023). Using this study design we propose to answer these research questions,

- RQ1 How do the learning outcomes, as measured by post-test scores across different task levels as defined by Bloom's taxonomy, differ between the control group and the experimental groups (with and without prompt training)?
- RQ2 How does the use of ChatGPT, with and without prompt training, impact the performance of students in data analysis tasks compared to a control group with no access to generative AI tools?

In this study, 157 learners (M = 117, F = 40) participated and all of them were first-year undergraduate students from an engineering college. The following measures were used - data literacy survey, pre-test, post-test, system usability scale, and additionally feedback on using prompts was obtained using experience surveys. The results indicate significant differences in post-test scores between the control group and experimental groups at different levels of Bloom's taxonomy and task performance, within experimental groups, experimental group 2 (G3) performed significantly better compared to group G2 which underscores the effect of prompt training.

The paper is structured as follows: The Literature Review section will provide an overview of existing research on the use of large language models in education and the theoretical framework of prompt training. The Methodology section will detail the experimental design, data collection measures, and analytical techniques employed in this study. The Results section will present the findings, highlighting the differences in performance across the control and experimental groups. The Discussion section will interpret the results, compare them with existing literature, and explore the implications for educational practice and future research. Finally, the Conclusion will summarize the key findings, discuss the limitations of the study, and suggest directions for future research.

2. Background

2.1. Generative AI in education

Generative AI employs machine and deep learning to autonomously create new data, marking a significant difference from traditional AI tasks (Feuerriegel et al., 2024). Large Language Models (LLMs) are at the core of this technology, which are constructed from extensive neural networks, and excel in processing and generating language-based data, producing outputs that closely mimic human creations (Brynjolfsson et al., 2023). Their underlying architectures, particularly the transformer model with self-attention mechanisms, enable LLMs to capture intricate linguistic patterns and relationships, facilitating advanced text generation (Hadi et al., 2023). The transformative potential of generative AI extends across multiple fields, including healthcare, business, research, and education (Fui-Hoon Nah et al., 2023). By leveraging its ability to generate high-quality content, these sectors have seen significant advancements in efficiency, creativity, and engagement. For example, in business, it enhances customer service through specific chatbots and personalized marketing strategies (Kshetri et al., 2023).

Generative AI has demonstrated applications in education, enhancing teaching and learning experiences. A few studies have highlighted its role in increasing student engagement and motivation. For instance, Lee et al. (2023) found that AI-based content generators improved reading enjoyment and interest among elementary students, making learning more engaging and interactive. Similarly, Guo et al. (2023) highlighted how AI-based chatbots can enhance task motivation and argumentation skills in undergraduate students, fostering critical thinking and effective communication. In a study by Aktay et al. (2023), fourth-grade students found ChatGPT engaging and beneficial for academic achievement, suggesting its use in various subjects. In the context of language learning, Jeon and Lee (2023) emphasized the complementary relationship between human teachers and ChatGPT, showcasing its role in providing personalized feedback and practice for language learners. Hong (2023) highlighted the opportunities ChatGPT presents in foreign language teaching, stressing the importance of ethical discussions with students about its use. Moreover, the broader impact of generative AI on educational methods and creative collaboration has been explored. Chen et al. (2022) reviewed two decades of AI in education, underscoring its potential to revolutionize instructional methods and student engagement. Hutson and Schnellmann (2023) discussed the collaborative role of AI in creating poetry, emphasizing the need to balance AI assistance and independent learning. These studies collectively illustrate the versatile applications of Generative AI in education, promoting personalized, efficient, and engaging learning experiences across different educational levels and subjects.

Extending beyond, generative AI has shown remarkable promise in the domain of programming also, recent studies have illuminated the impactful role of generative AI in this field. An experimental study showcased how the programming tool Codex, powered by generative AI, outperformed learners in a CS1 class on a rainfall problem, ranking in the top quartile (Denny et al., 2023). Another investigation used the flake8 tool to assess code generated by AI against the PEP8 coding style, revealing a minimal syntax error rate of 2.88% (Feng et al., 2023). A notable study involved Github's generative AI platform, which initially failed to solve 87 Python problems; however, applying prompt engineering techniques enabled it to resolve approximately 60.9% of them successfully (Finnie-Ansley et al., 2022). These findings collectively highlight the efficacy of generative AI in code generation. Despite these advancements, there is a notable gap in empirical research examining the influence of prompt training in the education domain. While prompt engineering has been shown to enhance the performance of generative AI in solving problems, its application and effectiveness in the context of data analysis education remain underexplored. This area needs further investigation to determine how targeted prompt training can improve learning outcomes and practical skills for students in data analysis courses.

2.2. Challenges in using GenAI

Alongside these advancements, generative AI also faces significant challenges. One major issue is hallucination in AI models, where the generated content deviates significantly from practical constraints or reality (Mukherjee & Chang, 2023). This problem arises because AI models, unlike traditional rule-based systems, learn from examples encountered during their training phase and lack an explicitly codified set of rules to separate fact from fiction (Varsha, 2023). Consequently, AI models may produce creative content that deviates too far from practical constraints (hallucination) or remains too rigidly within the confines of existing data (memorization) (Prasai, 2023) (Ji et al., 2023). Striking a balance between novelty and usefulness is crucial for generating original content that is both relevant and practically applicable in a given context (Mukherjee & Chang, 2023). Understanding and mitigating hallucination is essential for developing AI systems that generate trustworthy and practical content (Sovrano et al., 2023). Additionally, AI models, trained on vast datasets, generate content based on probabilities without understanding real-world contexts, leading to potentially misleading and irrelevant outputs (Ebert & Louridas, 2023). Ensuring the accuracy and reliability of AI-generated content is crucial.

Ways to address these challenges are included both from the designer's end and from the learners' end. Improvements by fine-tuning the model and development of human-AI collaboration are from the designers' perspective (Leiser et al., 2023). From the learner's end, one promising solution to these challenges can be prompt engineering. Prompt engineering extends the capabilities of large language models (LLMs) and vision-language models (VLMs) by providing task-specific instructions (Sahoo et al., 2024). These prompts guide model outputs without modifying core parameters, allowing seamless integration into diverse tasks (Sahoo et al., 2024) (Shah, 2024). This approach enhances model efficacy, enabling success in applications ranging from question-answering to commonsense reasoning. Prompting leverages pre-trained knowledge to generate desired behaviors, enhancing adaptability and reducing the need for extensive retraining (Kucharavy, 2024) (Sahoo et al., 2024) (Shah, 2024). By systematically categorizing various prompting techniques and highlighting their strengths and limitations, prompt engineering offers a practical way to address the challenges posed by generative AI (Brown et al., 2020) (Wei et al., 2022) (Radford et al., 2019).

2.3. Prompt engineering

The journey of prompt engineering begins with zero-shot prompting, where models perform tasks based on general instructions without specific training data, as demonstrated by Radford et al. (2019). Next, few-shot prompting introduces a few examples to improve model understanding, with Brown et al. (2020) showing that providing high-quality examples enhances performance on complex tasks. Chain-of-thought (CoT) prompting, introduced by Wei et al. (2022), advances the technique by enabling step-by-step reasoning, significantly improving performance in reasoning tasks. To automate this process, Zhang et al. (2022) proposed Automatic Chain-of-Thought (Auto-CoT) prompting, which generates diverse reasoning chains automatically, improving accuracy and robustness without manual effort. Further refining CoT prompting, Wang et al. (2022) introduced self-consistency, generating multiple reasoning chains and selecting the most consistent final answer to enhance accuracy through diverse approaches. Finally, Logical Chain-of-Thought (LogiCoT) prompting, proposed by Zhao et al. (2023), integrates symbolic logic principles to verify each reasoning step, reducing errors and hallucinations.

To capture the nuance of these prompting techniques, frameworks are designed to facilitate their practical implementation in real-world applications. Moreover, prompting frameworks play a crucial role in bridging the gap between the model's capabilities and the practical needs of users. They provide the necessary infrastructure for integrating external tools, maintaining historical information, and ensuring structured and safe outputs (Liu et al., 2023). For example, frameworks like LangChain and Semantic Kernel enable LLMs to interact with databases, web browsers, and other external systems, thus overcoming the LLMs' inherent limitations and extending their applicability (Liu et al., 2023). In this study we have chosen the CLEAR (Concise, Logical, Explicit, Adaptive, Reflective) framework as mentioned in Table 1, offering a systematic strategy for crafting prompts that harness the full potential of AI language models. It's not merely about the individual components of clarity, context, formatting, or verbosity control; it's about how these elements synergize through a comprehensive, adaptable framework to elevate the entire communication process with AI (Lo, 2023).

3. Methods

3.1. Experimental procedure

The study emphasizes two critical data analysis concepts: data aggregation and merging. Aggregation simplifies data, revealing trends and easing novices into complex tasks, much like learning the alphabet before forming sentences (McKinney, 2022). Merging integrates diverse datasets, essential for the data landscape, offering a unified view (McKinney, 2022). The tasks of data aggregation and merging that are to be completed are designed in a way that cannot be completed with non-programming software like Excel and Tableau. We have selected Python programming because of its relevance in real-world applications, along with market demand, as highlighted by Ying and Zhang (2019). The dataset includes dummy data of tablet usage by school students from September to January, capturing attributes such as student video usage, student ID, school ID, view count, and last access date and time. Each month's dataset contains over 10,000 observations, providing a comprehensive view of student engagement with video content. This dataset draws inspiration from a school education program where students are provided tablets to enhance learning. The following are the problem statements of the task:

- T1: Calculate the total daily video usage for each student across all months. (In the dataset, students watched the content multiple times a day. Participants need to calculate the total daily video usage for each day and repeat this task for all four months of the dataset October to January).
- T2: Given the unique data capture cycle of student video usage (the 26th of one month to the 25th of the next), compute the monthly total video usage for each student. For example, compute the total video usage for October (1st October to 31st October).

Table 1
Elements of the CLEAR framework provide guidance for writing structured prompt (Lo, 2023).

Characteristic	Description
Concise	Prompt should be clear and focused on the task's core elements to guide AI towards relevant and precise responses.
Logical	Prompts need a logical flow of ideas, helping AI understand context and relationships between concepts, resulting in coherent outputs.
Explicit	Prompts must clearly specify the expected output format, content, or scope to avoid irrelevant responses.
Adaptive	Prompts should be flexible, allowing experimentation with different structures and settings to balance creativity and specificity.
Reflective	Continuous evaluation and refinement of prompts are crucial, using insights from previous responses to improve future interactions.

- T3: Calculate the monthly video usage for each school over all the months. (Each school has multiple students, so aggregate the total video usage for each month and compute the total video usage during the intervention period for each school).

Datasets and tasks are provided to students, and the output CSV/Excel file is collected at the end of the task.

3.2. Participants

We grouped learners into three groups - the control group (access to the internet but no access to GenAI), experimental group 1 (access to the internet and GenAI without prompt training), and experimental group 2 (access to the internet and GenAI with prompt training). In total 157 learners participated in this study all were first-year undergraduate students from an engineering college. Though the students were from different streams all of them are first-year undergraduate students for whom the curriculum remains the same regardless of their specialization. The stream-specific courses start from the second year onwards.

The sample size is predicated upon several critical metrics, namely effect size, power, and alpha level. According to the methodologies outlined by Endres et al. (2017), an effect size of 0.35 was selected, representing the expected magnitude of differences between the groups. To achieve a statistically robust analysis, a power level of 0.8 was established. This power level signifies an 80% probability of detecting true effects, thereby minimizing the risk of Type II errors (Cohen, 2013). An alpha level of 0.05 was set to control for the likelihood of Type I errors, ensuring the precision and accuracy of the results. Utilizing these parameters, a GPower analysis was conducted to determine the required sample size for the study. GPower, a widely recognized tool for power analysis, indicated that a sample size of 125 participants would be necessary to achieve reliable results using the one-way ANOVA technique.

To facilitate effective comparative analysis, the study flow for various groups is shown in Fig. 1. A survey on familiarity with the use of GenAI was collected followed by pre-test, tasks, and post-test. Prompt training was conducted for group 3 before the start of the task. Ethical approval was obtained from the Institute Review Board of the authors' institution.

3.3. Measures

3.3.1. Data literacy survey

In line with Hamilton et al., who define data literacy as "the ability to ask and answer questions about collecting, analyzing, and making sense of data," a customized four-item Likert-scale survey was developed to evaluate participants' competence in using generative AI for data analysis (Hamilton et al., 2009). This measure encompasses two primary dimensions: (1) confidence in applying AI-driven solutions, and (2) frequency of AI usage in practical data-analysis scenarios. The scale design drew on several sources in the existing literature. Oguguo et al. highlighted critical dimensions of student data literacy, informing the survey's emphasis on tangible skill assessment (Oguguo et al., 2020). Zheng provided methodological insights into teaching and learning data analytics, which guided item development regarding practical AI usage (Zheng, 2019). Finally, da Silveira et al. underscored the importance of programming languages in cultivating analytical proficiency, shaping the survey's focus on technical fluency (Balreira et al., 2023). Together, these references helped ensure that the survey items captured both the theoretical underpinnings of data literacy and its real-world application within generative AI contexts.

Additionally, the data literacy survey questionnaire was meticulously reviewed by experts to ensure alignment with the curriculum requirements of engineering colleges in India. Further statistical evaluations, including a Cronbach's alpha value of 0.71, confirmed the questionnaire's reliability, making it a robust tool for assessing data literacy skills within this educational context.

3.3.2. Pre-test and post-test of domain knowledge

A set of 15 multiple choice questions was used for the pre-test and post-test focusing on the concepts of aggregation and merging. These questions assessed the participants' comprehensive understanding of the concepts through Python programming. Designed across three levels of Bloom's taxonomy - understanding (L1), applying (L2) and analyzing (L3), the test comprised five questions each for each level. The questions were adapted to assess concepts of data aggregation and data merging in Python, drawing inspiration from the official Pandas documentation and the book Python for Data Analysis by McKinney (2022).

- Item Development and Expert Review: Each item underwent face- and content-validity checks with domain experts (academics and industry professionals). We refined or removed ambiguous items during the development phase based on expert feedback. This review ensured that questions accurately assessed data aggregation and merging skills at the intended Bloom's taxonomy level.
- Statistical Item Analysis: We conducted item analysis on the pre-test scores to evaluate question quality. Cronbach's alpha was 0.42 for the overall test, while modest, was deemed acceptable given the exploratory nature and specific context of this study. While this value is below the conventional benchmark of 0.70, we believe it can be partly attributed to:
  - The short test length (15 items split across three distinct cognitive levels).
  - The multidimensional nature of the instrument (covering understanding, application, and analysis).

Despite this modest alpha, the iterative review process ensures construct and content validity in line with our educational objectives.

3.3.3. Task score

The tasks mentioned in section were evaluated using a rubric. Task completion scores were calculated across three tasks, including the total score. For Task 1 and Task 2, each task is worth a maximum of 4

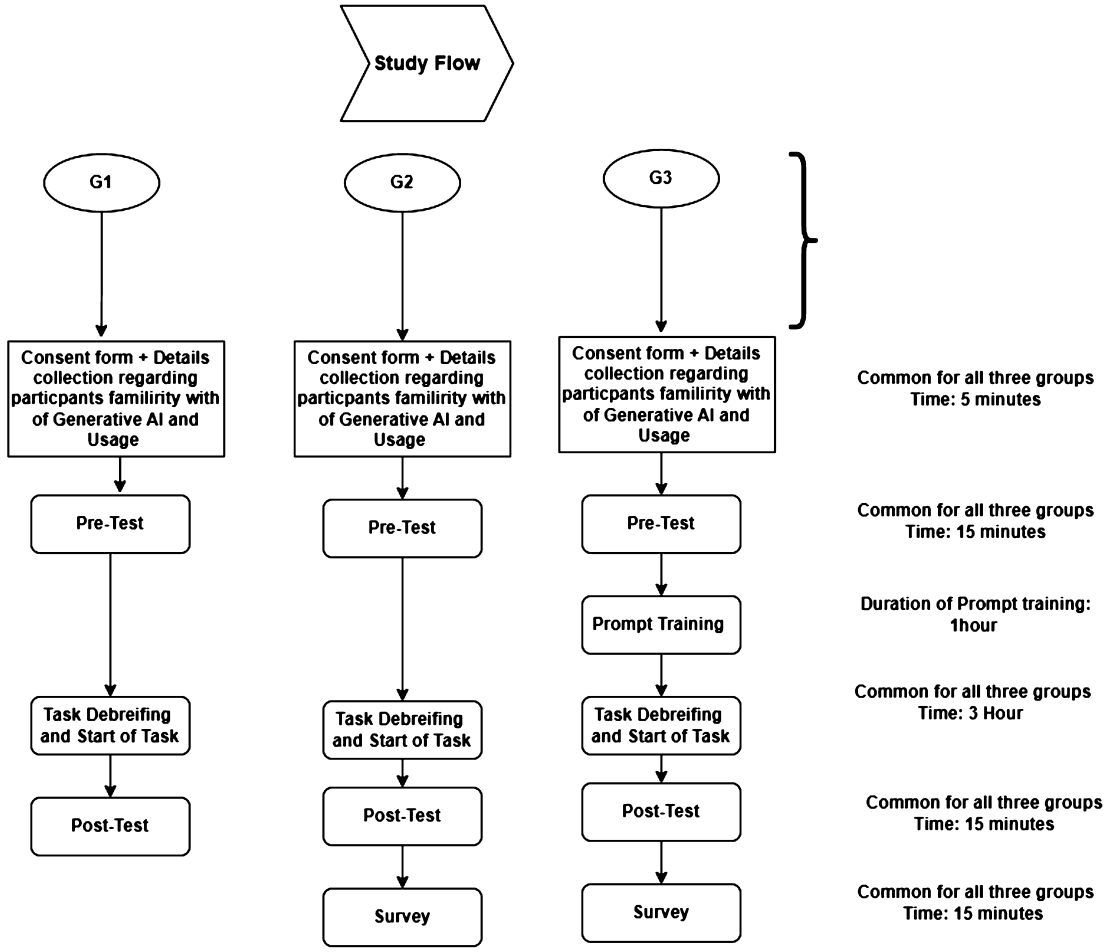


Fig. 1. The figure explains the study flow for all three groups and what the difference in the process.

points, where students earn 1 point per correctly produced data output file for each month, as specified in the dataset. Thus, students who generate all four required files correctly receive the full 4 points for these tasks. Task 3, however, has a maximum score of 2 points and requires the correct merging of data across all schools. Here, students receive 1 point for a correctly merged dataset but lose 1 point if any school is missing from the merged data, reflecting the importance of comprehensive data integration. Overall, the scoring system emphasizes accuracy in file generation and completeness in data merging.

3.3.4. System usability scale

The System Usability Scale (SUS) was adapted to capture feedback on the Generative AI tool used in data analysis. This scale is a widely utilized 10-item questionnaire that has been validated across numerous domains to measure perceived ease of use and user satisfaction. The SUS uses a balanced mix of positive and negative items scored on a 5-point scale, suitable for evaluating the tool's effectiveness and user experience. A comprehensive evaluation of the tool's impact on user proficiency can be achieved by triangulating SUS scores with task performance and learning metrics (Brooke, 1996).

3.4. Experience survey

The Experience Survey was conducted to gather detailed feedback from participants regarding their use of prompts in data analysis tasks. The survey focused on the usability of prompts and suggestions for improvement. The feedback collected through this survey was analyzed to identify common themes and areas for improvement.

3.5. Prompt training

In the previous detailing of the control group and experimental group, it is highlighted that one subset of participants was granted access to Generative AI and underwent training in the prompt formulation. To facilitate this, a one-hour instructional session was organized before the main task as depicted in Fig. 2. During this session, participants are introduced to a handout that explains the fundamentals of generative AI and various prompting techniques such as zero prompting, few-shot prompting, and chain prompting. It also includes the CLEAR framework for prompting and strategies from OpenAI. Students read through the handout to understand these basic concepts, followed by a short group discussion to assess their understanding and address any questions they may have.

Then, students apply their knowledge by writing a prompt for a given question mentioned in Fig. 3. This exercise allows them to reflect on their understanding and practice their prompt-writing skills. After five minutes, we present a structured prompt for the same question and facilitate a discussion to analyze its structure and the concepts employed. This helps students connect the learned concepts to the prompt, reinforcing their comprehension.

In the next phase, students work on a data analysis problem mentioned in Fig. 4 using a provided dataset that is different from the task dataset. They write a prompt to generate Python code for the analysis, applying their previous learning and seeking assistance online or from ChatGPT if needed. After their attempt, we present a structured prompt for the same problem and engage in a discussion to decode and explain the strategies used. Students then use the prompt to generate and run the Python code, ensuring it functions as expected.

Table 2
Establishing Baseline Homogeneity on the basis of Pre-test score of participants.

Category	G1 (N = 64)	G2 (N = 34)	G3 (N = 59)	ANOVA/χ^2 p-value
Male	46	29	42	
Female	18	5	17	$\chi^2(2) = 2.66$, p = 0.2645
Data Literacy Survey				
Confidence in performing data analysis tasks	2.53 (1.08)	2.53 (0.98)	2.73 (0.86)	F(2, 154) = 0.742, p = 0.4777
Confidence in programming with Python	2.64 (0.87)	2.38 (0.67)	2.54 (0.60)	F(2, 154) = 1.026, p = 0.3610
Frequency of using generative AI for data analysis	3.09 (1.20)	2.65 (1.08)	3.15 (1.17)	F(2, 154) = 2.632, p = 0.0751
Confidence in using generative AI for data analysis	3.08 (1.03)	2.50 (0.86)	2.93 (0.75)	F(2, 154) = 4.238, p = 0.0899
Pre-test Scores (Mean ± S.D)				
Total Score	3.85 (2.50)	3.47 (2.71)	3.83 (2.14)	F(2, 144) = 0.736, p = 0.481

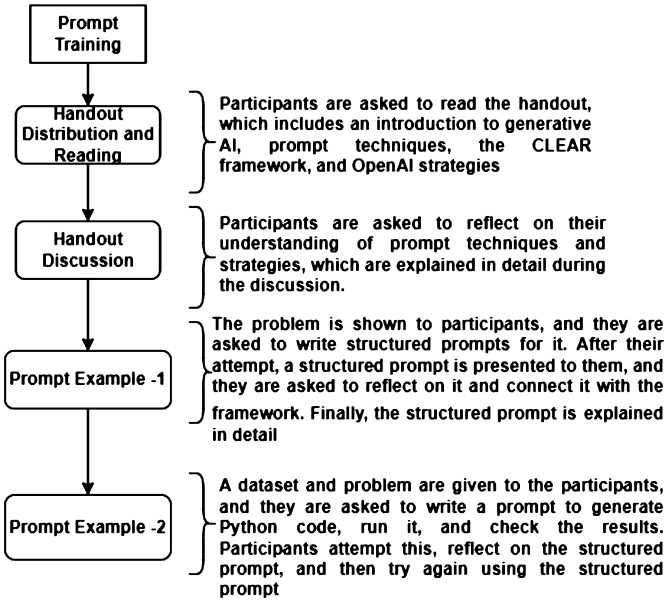


Fig. 2. The flowchart outlines the structured approach used to train participants in writing effective prompts, incorporating generative AI, prompt techniques, the CLEAR framework, and OpenAI strategies and examples.

This instructional design ensures that students not only grasp the theoretical aspects of prompting but also apply them practically, creating a comprehensive learning experience. By integrating reading, discussion, application, and reflection, we foster a holistic learning environment that promotes deeper understanding and skill development in prompt writing and generative AI.

To ensure that no undue advantage was conferred in the task domain, the live example provided was unrelated to data analysis tasks. It is worth noting that, post-study, the remaining two groups were also educated on prompt formulation to uphold ethical standards.

3.6. Statistical analysis

All statistical analyses were performed utilizing the Real Stats add-on for MS Excel. To compare data literacy scores among the three groups (one control and two experimental), a one-way Analysis of Variance (ANOVA) was employed, as it is suitable for assessing mean differences across multiple independent groups. Gender-based effects were analyzed using a chi-square test appropriate for examining associations between categorical variables. The chi-square assumptions were satisfied because the observations were independent, the sample size exceeded 20, expected frequencies were at least 5 in every category, and sampling was random.

To ensure group homogeneity before evaluating treatment effects, pre-test scores were compared across groups using ANOVA. The impact of prompt training and generative AI usage on outcomes was also analyzed through ANOVA, allowing for the assessment of significant intervention effects. Although standard ANOVA is generally robust to moderate violations of the homogeneity-of-variance assumption, Welch's ANOVA provides an additional safeguard when group sizes and/or variances differ significantly. For Welch's ANOVA and ANOVA, the normality assumption was assessed using the Shapiro-Wilk test ($p > 0.05$), indicating no deviation from normality. The variance assumption was checked using Levene's test ($p > 0.05$), confirming the homogeneity of variances.

Post-hoc comparisons were conducted using Tukey's Honestly Significant Difference (HSD) test to identify specific differences between group means. The assumptions for Tukey HSD were met because the data satisfied ANOVA assumptions for normality ($p > 0.05$ via Shapiro-Wilk test) and homogeneity of variances ($p > 0.05$ via Levene's test), with independent observations and a significant ANOVA result ($p < 0.05$). To investigate the relationship between programming knowledge and data analysis task performance, Hake gain analysis was performed to measure improvements in understanding. Non-linear correlations were then explored using Spearman's rank correlation coefficient to assess the relationship between Hake gain scores and task completion performance.

4. Results

4.1. Baseline homogenity

Baseline homogeneity was assessed among all 157 participants using a data literacy survey, demographic analysis, and pre-test scores. All participants were first-year engineering students in India, aged between 18 and 20, with a common curriculum background. Group details are provided in Table 2. The survey included Likert scale-based questions designed to gauge participants' skills in data analysis and programming. The responses were recorded on a 1-5 scale, with 1 indicating low confidence or frequency and 5 indicating high confidence or frequency. ANOVA was conducted for each question to test for significant differences between the groups. The results, including means, standard deviations, and p-values, are presented in Table 2. No significant differences were found among the groups for any question ($p > 0.05$), confirming the homogeneity in baseline skills. Additionally, the chi-square test confirmed no significant gender-based effect, $\chi^2(2) = 2.66$, p = 0.2645.

These results indicate no significant differences among the groups in their pre-test scores for any level ($p > 0.05$), confirming baseline homogeneity in programming skills related to data analysis. This ensures that any observed effects in the main study are due to the experimental conditions rather than pre-existing differences among the participants.

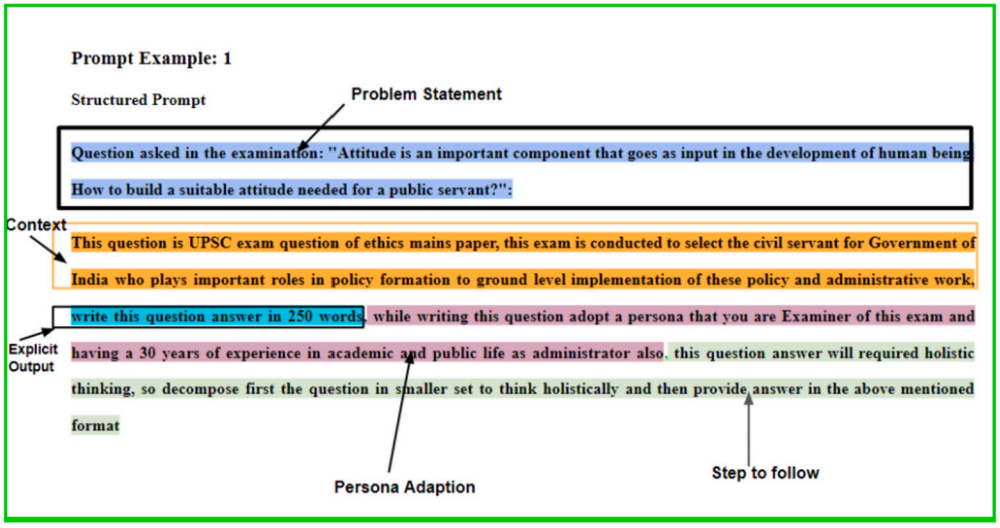


Fig. 3. Structured prompt example -1, explain the concept and strategies mentioned to write the prompt, this example is for the UPSC problem.

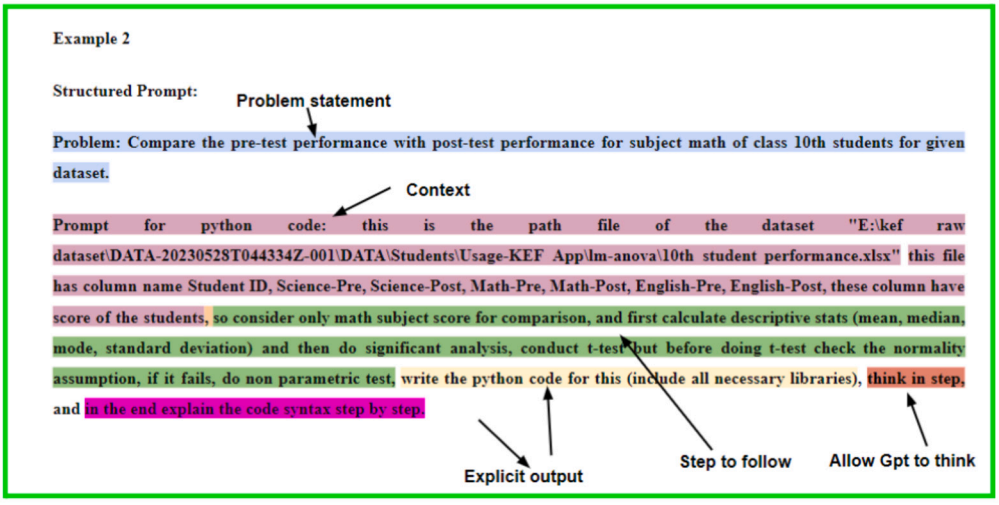


Fig. 4. The Figure explains for structured prompt for a given dataset problem to generate Python code and asks to explain the code.

4.2. Effect of prompt training and generative AI on programming learning

The impact of the intervention on participants' programming knowledge was analyzed using ANOVA on post-test scores for understanding (L1), application (L2), and analysis (L3) levels, as well as the total score mentioned in Table 3. The table presents the results of 144 students who completed the post-test. Out of the 157 students who consented to participate in the study, 13 did not take the post-test. The results showed significant differences among the groups for all levels, indicating a substantial impact of the intervention. G3 consistently outperformed G2 and G1, with the highest mean scores across all levels. For L1 (understanding), G3 scored 2.69 (s.d = 0.88), G2 scored 2.31 (s.d = 0.87), and G1 scored 1.39 (s.d = 0.35). Similar trends were observed for L2 (application) and L3 (analysis), with G3 scoring significantly higher than the other groups. The total post-test scores also revealed G3's superior performance (6.60, s.d = 2.69) compared to G2 (4.94, s.d = 2.25) and G1 (4.28, s.d = 1.17). These results underscore the effectiveness of the intervention prompt training with the integration of generative AI, particularly for G3.

Further analysis using Tukey's HSD test revealed significant differences between specific groups. These findings highlight the effectiveness of the intervention for G3, suggesting that the training program significantly enhanced their programming skills in data analysis compared to G2 and G3. Additionally, at understanding (L1) G2 performed substantially better compared to G1, highlighting the effectiveness of generative AI compared to internet search. At the same time, performance at level 2 and level 3 is not significant which highlights the need for prompting skills.

4.3. Impact of intervention on task completion scores

To evaluate the impact of the intervention, we analyzed the task completion scores across three tasks and the total score for all 157 participants. Each task was rated based on a detailed rubric, with scores out of 10. The ANOVA results in Table 4 indicate significant differences among the groups for all tasks, highlighting the intervention's effect. G3 consistently scored highest across all tasks and the total score. For Task 1, G3 scored 3.41 (s.d = 0.75), significantly higher than G2 (2.74, s.d = 0.83) and G1 (1.11, s.d = 0.94), with $F(2, 154) = 117.38$, $p < 0.001$. A similar trend was observed for Task 2, with G3 outperforming G2 and G1. For task 3, G2 and G3 outperformed G1, but there was no significant difference between G2 and G3. The total score analysis also showed significant differences, with G3 scoring 8.76 (s.d = 1.43) compared to G2 (6.91, s.d = 2.34) and G1 (2.00, s.d = 2.26), with $F(2, 154) = 183.15$, $p < 0.001$.

Table 3
Comparison of Post-Test Scores Using ANOVA, Welch's ANOVA and Post Hoc Test, Indicating Significant Differences in Post-Test Scores at All Levels and Total Score Among the Three Groups.

Level	G1 (Mean ± S.D)	G2 (Mean ± S.D)	G3 (Mean ± S.D)	ANOVA F(df), p-value	Welch's ANOVA F' (df1,df2), p-value	G1 vs G3 (p-value)	G2 vs G1 (p-value)	G3 vs G2 (p-value)
L1: Understanding	1.39 (0.35)	2.31 (0.87)	2.69 (0.88)	F(2, 141) = 37.22, **1.05E-13**	43.075 (2,71.76), **5.13E-13**	**8.92E-14**	**2.76E-06**	0.097
L2: Application	1.56 (0.46)	1.38 (0.44)	2.24 (0.41)	F(2, 141) = 22.28, **3.90E-09**	22.8298 (2,80.53), **1.40E-08**	**7.91E-07**	0.410	**9.01E-08**
L3: Analysis	1.32 (0.40)	1.28 (0.34)	1.84 (0.44)	F(2, 141) = 12.12, **1.39E-05**	11.7119 (2,82.99), **3.31E-05**	**7.42E-05**	0.967	**3.59E-04**
Total Score	4.28 (1.17)	4.94 (2.25)	6.60 (2.69)	F(2, 141) = 39.29, **2.75E-14**	38.4089 (2,74.41), **3.48E-12**	**1.48E-14**	0.092	**1.30E-06**

Table 4
Tasks scores across groups G1, G2, and G3.

Level	G1 (Mean ± S.D)	G2 (Mean ± S.D)	G3 (Mean ± S.D)	ANOVA F(df), p-value	Welch's ANOVA F' (df1,df2), p-value	G1 vs G3 (p-value)	G2 vs G1 (p-value)	G3 vs G2 (p-value)
Task 1	1.11 (0.94)	2.74 (0.83)	3.41 (0.75)	F(2, 154) = 117.38, **1.08E-31**	113.31 (2, 86.37), **8.63E-26**	**1.69E-14**	**1.51E-14**	**9.62E-04**
Task 2	0.75 (1.57)	2.59 (1.94)	3.66 (1.12)	F(2, 154) = 57.81, **1.87E-19**	69.91 (2, 77.23), **7.29E-18**	**1.69E-14**	**1.63E-07**	**3.55E-03**
Task 3	0.28 (0.70)	1.59 (0.82)	1.69 (0.73)	F(2, 154) = 66.15, **1.84E-21**	68.59 (2, 82.76), **2.07E-21**	**1.69E-14**	**9.02E-14**	0.78
Total Score	2.00 (2.26)	6.91 (2.34)	8.76 (1.43)	F(2, 154) = 183.15, **1.94E-41**	198.05 (2, 77.97), **6.75E-47**	**1.69E-14**	**1.69E-14**	**9.69E-05**

Table 5
Hake gain of pre and post-test, total task score, and Spearman correlation for groups G1, G2, and G3.

Group	Hake Gain (Mean ± S.D)	Total Task Score (Mean ± S.D)	Spearman Correlation
G1	0.0401 ± 0.2609	2.00 ± 2.26	r(42) = 0.72, P value = **4.48 E-08**
G2	0.1274 ± 0.0878	6.91 ± 2.34	r(25) = 0.90, P value = **1.42 E-10**
G3	0.2463 ± 0.0044	8.76 ± 1.43	r(48) = 0.77, P value = **2.93 E-12**

Table 4 shows the mean and standard deviation (Mean ± S.D) of post-test scores for tasks and total scores across groups G1, G2, and G3. ANOVA results indicate significant differences, with post hoc test p-values highlighting specific group differences.

To further understand the relationship between programming knowledge and data analysis task performance, we conducted a Hake Gain analysis and non-linear correlation analysis (Spearman's correlation) between the Hake Gain scores and data analysis task completion scores. The Hake Gain analysis results are summarized in Table 5

Table 5 presents the Hake gain of pre and post-test, total task score, and Spearman correlation for groups G1, G2, and G3. Of the 157 participants, 121 completed both the pre- and post-tests as well as the task. The correlation coefficients (r) and p-values indicate the strength and significance of the relationship between Hake gain and total task scores within each group.

There is a strong positive correlation between programming knowledge and data analysis task performance across all groups. Group 3 scored high in both areas, demonstrating a strong association between programming knowledge and better performance in data analysis tasks. Group 2 showed a significant role of programming knowledge in their intermediate performance levels, while Group 1, despite having lower scores, also displayed a strong association between programming knowledge and task performance. These findings emphasize that programming knowledge is a critical factor in enhancing data analysis task performance across all groups

4.4. System usability scale results

To evaluate the usability of ChatGPT, a comparison of System Usability Scale (SUS) scores was conducted between experimental group 1 (G2), which did not receive prompt training, and experimental Group 2 (G3), which received prompt training. An independent samples t-test assuming equal variances revealed a statistically significant difference in SUS scores between the two groups, with G3 reporting a mean SUS score of 77.33 (SD = 9.64) and G2 a mean score of 73.16 (SD = 10.18), t(91)

= 1.967, p = 0.026 (one-tailed). The effect size (r = 0.202) indicates a small but meaningful impact of prompt training on usability. These results suggest that prompt training significantly improves users' perceived usability of ChatGPT, underscoring its importance for enhancing user experience.

5. Discussion

5.1. Effect of generative AI and prompt training in programming knowledge

To address RQ1, "What is the impact of integrating Generative AI with prompt engineering on first-year engineering students' learning in the domain of programming for data analysis?" we compared the performance of different groups. Specifically, Group 3 (G3), which utilized both Generative AI and prompt engineering, outperformed Group 1 (G1), which relied on traditional internet searches, across all three levels of programming knowledge. This demonstrates the effectiveness of the combined approach of Generative AI and prompt engineering over traditional methods. However, when examining the effect of Generative AI alone, the results indicate significant improvement only at the understanding level, but not at the other two levels. This finding suggests that while Generative AI alone contributes to learning, its full potential is realized when combined with structured prompt engineering.

To verify the role of prompt engineering, we compared Group 2 (G2), which used Generative AI without prompt training, to Group 3 (G3). The comparison revealed that no significant differences emerged at the understanding level, which is the lower-order thinking skills of Bloom's taxonomy. Nonetheless, the overall scores and performance at higher cognitive levels were significantly better in G3, highlighting the critical role of prompting skills in enhancing programming learning. These findings align with several recent studies emphasizing the role of Generative AI as a learning tool and the efficacy of prompting techniques. The enhanced learner performance observed in our study can be further explained by the higher System Usability Scale (SUS) scores reported by participants in G3 compared to G2. A more interactive and supportive environment with high usability reduces cognitive load, allowing participants to focus more on domain learning. Participants reported that prompt techniques, which involved asking the AI to explain the syntax and underlying concepts of generated code, significantly aided their understanding of programming intricacies compared to both control (G1) and experimental groups (G2,G3).

Our findings that integrating Generative AI with prompt engineering significantly enhances first-year engineering students' learning outcomes in programming for data analysis are corroborated by few recent studies. For instance, Chan and Hu (2023) observed that ChatGPT's ability to handle repetitive tasks allows students to focus on advanced learning, enhancing their performance. Similarly, Alneyadi and Wardat (2023) demonstrated a clear improvement in post-test scores for students using ChatGPT, highlighting its effectiveness in educational settings. Additionally, Boubker (2024) found that personalized approaches with ChatGPT improve students' perceived usefulness and learning effectiveness, aligning with our results that structured prompt training enhances understanding and mastery of programming concepts. These studies collectively support our conclusion that combining Generative AI with prompt engineering is more effective than using traditional methods or AI alone.

5.2. Effect of generative AI and prompt training in data analysis task completion

To address RQ2, "What is the impact of integrating Generative AI with prompt engineering on first-year engineering students in data analysis task completion?" we analyzed the performance of different groups in the given tasks. We found that Group 3 (G3), which utilized both Generative AI and prompt engineering, consistently outperformed all other groups. The task scores for G3 were significantly higher across all three individual tasks and the total task score level, underscoring the combined benefits of Generative AI and prompt techniques. This was contrasted with Group 1 (G1), which relied solely on traditional methods, and Group 2 (G2), which used Generative AI without prompt training. When comparing G2 to G1, we found significant differences in all task scores, both individually and in total, indicating that even without prompt training, Generative AI substantially improves task completion rates compared to traditional methods. This finding aligns with previous research in programming, where code generated by AI tools often surpasses the quality of code written by novice learners within a limited time frame. Moreover, the higher System Usability Scale (SUS) scores reported by participants in G3 further support these results, indicating a more user-friendly and supportive learning environment. Recent studies show that accuracy enhancements of up to 13.79% have been observed in code generation tasks Li et al. (2023), and a 12% gain in complex reasoning tasks was achieved Li et al. (2023). These improvements support our findings that integrating Generative AI with prompt engineering significantly enhances first-year engineering students' task completion in data analysis, highlighting the effectiveness of structured prompting in educational settings.

Additionally, a Spearman's correlation analysis revealed a strong positive correlation (approximately 0.8) between programming knowledge and task completion scores, suggesting that participants who performed well in tasks also showed improved programming knowledge. This indicates that while Generative AI facilitates task completion, the retention and deeper understanding of programming concepts develop over time. Our study demonstrates that integrating Generative AI with prompt engineering significantly improves data analysis task completion among first-year engineering students. This integration not only enhances immediate task performance but also supports deeper learning and retention of programming concepts, making it a valuable tool in educational settings.

5.3. Experience survey feedback

The Learning Experience Survey results provide additional support for the efficacy of integrating Generative AI with prompt engineering. Participants reported that diverse explanations offered by AI catered to various learning styles, fostering a deeper understanding of programming concepts. The structured and iterative nature of prompt crafting promoted reflective learning and skill reinforcement. Additionally, the interactive and immediate feedback from AI-fueled curiosity and encouraged continuous discovery. These qualitative insights corroborate our quantitative findings, demonstrating that prompt engineering, combined with Generative AI, not only improves performance in programming tasks but also enriches the overall learning experience.

Additionally, participants noted that prompt training improved their questioning skills, emphasizing the importance of context in generating code, breaking down problems, and logically structuring queries for better responses. This suggests that prompt engineering not only enhances the effectiveness of Generative AI tools but also boosts learners' efficiency and problem-solving abilities during experiential learning.

5.4. Implications for educational theories and pedagogy

Our results show that structured prompt training (G3) fosters gains not only at the lower levels of Bloom's taxonomy (understanding) but also at higher levels (application and analysis). By teaching students how to formulate effective prompts and interpret AI-generated feedback, educators can scaffold progression across cognitive domains, moving learners from basic comprehension toward more complex analytical and evaluative tasks. This finding underscores the value of explicit prompt-engineering instruction in helping novice programmers advance along Bloom's hierarchy more rapidly and effectively.

From a constructionist perspective, the prompt-training process functions as an effective scaffold, guiding students through a cycle of experi-

mentation, feedback, and reflection. Rather than passively receiving information, learners actively construct their programming knowledge by crafting prompts, testing AI responses, and refining their understanding of both code logic and data-analysis techniques. This iterative process echoes the hallmarks of cognitive apprenticeship, where novices learn expert strategies (e.g., chaining prompts, using the CLEAR framework) through guided practice. Integrating these scaffolds into a programming curriculum can thus enhance engagement and accelerate the development of problem-solving skills.

Lastly, blending Generative AI with structured prompt training closely aligns with 21st-century educational objectives that emphasize critical thinking, digital fluency, and adaptable problem-solving. By interacting with AI-driven tools, students in G3 cultivated planning queries, evaluating AI outputs, and iterating solutions—all of which are indispensable for modern engineering roles. Moreover, when basic or repetitive coding tasks are offloaded to AI, instructors can reorient class time to deeper conceptual discussions and collaborative projects, fostering the creativity, communication, and self-directed inquiry advocated by contemporary learning frameworks.

In practice, these findings suggest that instructors should consider incorporating explicit prompt-engineering lessons—covering both technical (e.g., how to specify parameters or contexts) and cognitive (e.g., how to break down a complex task into smaller prompts) dimensions—into early programming curricula. Doing so can help students leverage AI more effectively for task completion and skill-building, while also honing their ability to articulate, analyze, and refine problem-solving strategies. Ultimately, by combining Generative AI with thoughtfully designed scaffolds, educators can foster robust learning environments that promote immediate task success.

6. Conclusion and future work

6.1. Conclusion

The integration of generative AI in education is transforming teaching and learning methodologies. This study explores the potential of structured prompt training to enhance the educational utility of AI-generated content. Our findings demonstrate that prompt training significantly improves learning outcomes, engagement, and the quality of AI interactions. In data analysis and programming tasks, participants who received structured prompt training outperformed those who did not receive prompt training. By delivering context-specific cues and structured guidance, prompt training significantly improves problem-solving skills by addressing the limitations of AI-generated responses, which often lack contextual relevance.

The study has the following limitations. The gender imbalance among participants and the lack of comparative gender analysis limit the generalizability of the findings. Additionally, the short duration of the study poses challenges in assessing the long-term effects of the interventions. The lack of detailed interaction analysis and quantification of prompting skills during task execution makes it difficult to understand the intricacies of the learning process.

6.2. Future work

Future research should address the limitations identified in this study to enhance our understanding of the impact of generative AI in education. Longitudinal studies are essential to explore the sustained impact of structured prompt training on learning outcomes over time. A more balanced gender representation and comparative gender analysis are necessary to ensure the results are applicable across different demographics. Multi-modal analysis, incorporating visual and auditory cues, should be pursued to gain deeper insights into student-AI interactions and cognitive processes. This approach can lead to more personalized and effective educational interventions. Developing sophisticated interaction analysis systems will be key in providing real-time feedback and adapting to individual learning styles, thereby optimizing learning outcomes. Theoretical advancements are needed to refine existing frameworks and guide the design of AI-driven educational interventions. New approaches to prompt engineering and user interface design can enhance the usability and effectiveness of AI tools in the learning-teaching domain.

CRediT authorship contribution statement

Ashish Garg: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **K. Nisumba Soodhani:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Ramkumar Rajendran:** Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Conceptualization.

Ethical considerations and data availability

The study was approved by the Institution Review Board ethical committee with ID: IIT-IRB/2021/006. Informed consent was obtained from all participants, and their privacy rights were strictly observed. The data can be obtained by sending request emails to the corresponding author.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT 3.5 in order to check the grammatical errors and improve readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We acknowledge SBI Foundation Hub for Data Science & Analytics, IIT Bombay for supporting the work done in this project. We thank the teachers and students who participated in this study, whose contributions were invaluable to this work.

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